



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SHREE PETROLEUM AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

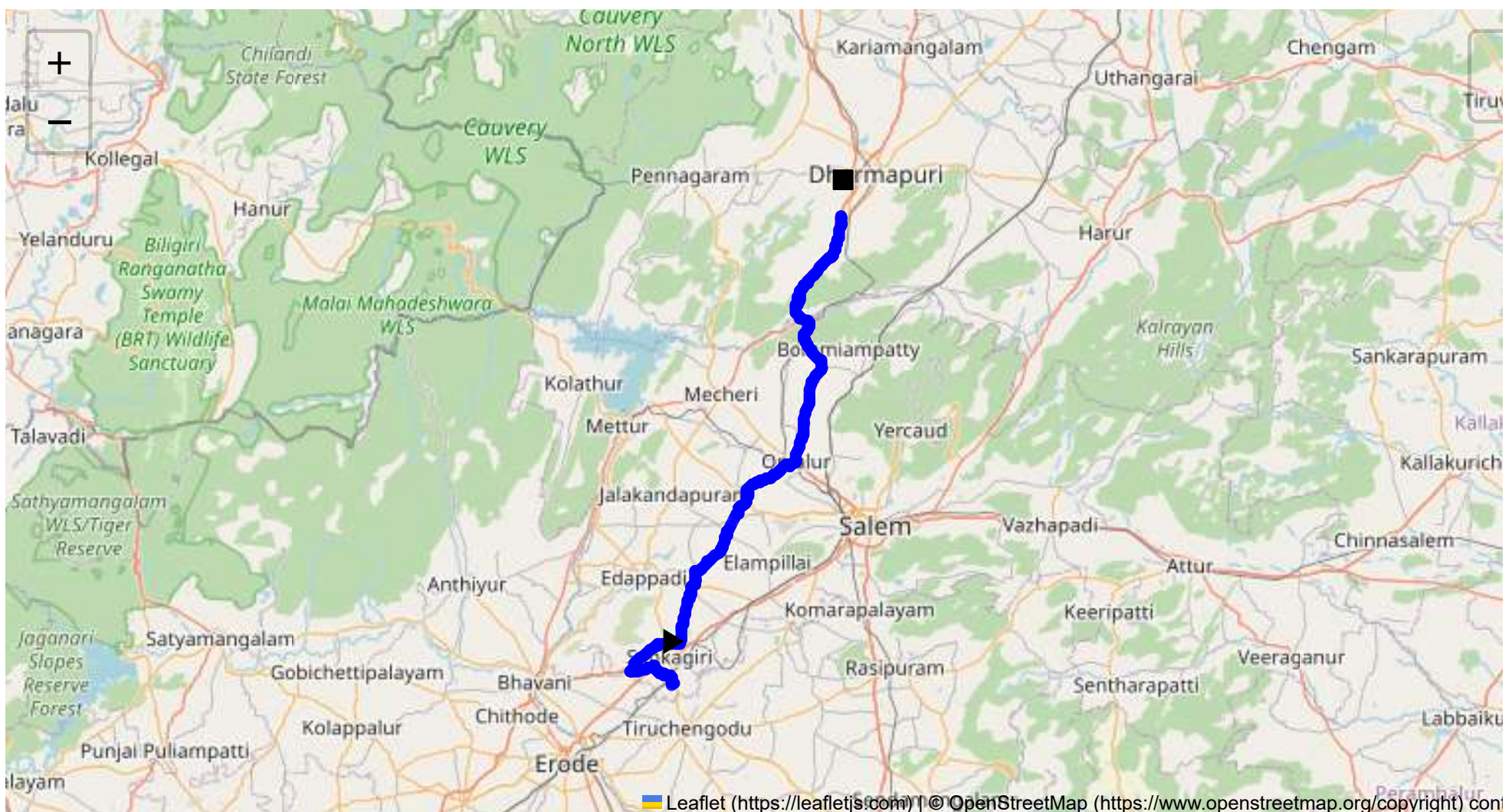
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 97.87 km**  
**Estimated Duration: 2.1 hours**  
**Adjusted Duration (Heavy Vehicle): 2.6 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.066068, 78.109762)**



## Welcome to the Journey Risk Management Study

### Route Overview

The route from CVQF+23W, Sangagiri to 3485+CW, Nallampalli covers approximately 97.87 kilometers and runs through several towns and urban areas in Tamil Nadu, India. The journey primarily uses state highways and some local roads, featuring varied traffic densities and infrastructure quality.



## Weather Conditions and Hazards

- **Typical Weather:** Tamil Nadu experiences a tropical climate. Summers can be extremely hot, with temperatures often exceeding 35°C, while the monsoon season, from June to September, brings heavy rain which may cause localized flooding.
- **Weather-related Hazards:** Monsoon rains can lead to slippery roads and reduced visibility. Strong winds during storms may affect high-sided vehicles.

## Traffic Patterns

- **Peak Hours:** Peak traffic is typically seen during morning (8 AM to 10 AM) and evening (5 PM to 8 PM) hours, especially in urban centers.
- **Congestion-prone Areas:** Traffic congestion is likely around major intersections, markets in towns such as Salem, and near industrial zones.

## Road Quality and Infrastructure

- **Road Conditions:** While the highways like the Salem - Kochi Highway are well-maintained, certain local roads and rural stretches may have potholes or uneven surfaces.
- **Infrastructure:** Bridges and overpasses are generally in good condition, but watch for narrow stretches and abrupt speed bumps in villages.

## Alternative Routes

For emergencies, alternative routes might include:

- Bypassing congested areas via the NH44 if new diversions arise.
- Consultation with local authorities for real-time detours during roadwork or blockages.

## Local Regulations

- **Hazardous Materials:** Restrictions on the times of day when heavy and hazardous material-laden vehicles can pass through certain areas, often outside of peak hours, need to be adhered to.

## Historical Incidents

- **Incidents:** There have been occasional accidents involving heavy vehicles, often due to speeding or mechanical failures. Specific data on hazardous material incidents may not be readily available but should be checked with local transport departments.

## Environmental Considerations

- **Sensitive Areas:** Areas near forests or water bodies (e.g., Cauvery river) require cautious driving to prevent environmental impact from potential spills.
- **Wildlife Crossings:** Close to countryside areas, be cautious of wildlife crossings, especially at night.

# Communication Coverage

- **Coverage:** Most of the route has good mobile network coverage, though certain rural or hilly areas may experience poor signal quality.
- **Potential Dead Zones:** Watch for signal loss in rural stretches between towns like Padaiveedu and Erumaipatti.

# Emergency Response Times

- **Estimated Response:** In urban areas like Salem, response times can be around 15-30 minutes due to proximity to emergency services. Response times may extend to 45-60 minutes in rural sections.

# Overall Summary of Risk Assessment

This route presents typical risks associated with rural and semi-urban Indian roads, including variable road quality and weather-related hazards. Congestion in urban areas and local regulations add complexity for hazardous material transport. Adequate preparation, understanding of alternate routes, and adherence to regulations are crucial for minimizing risks. Moreover, continuous monitoring for real-time traffic updates and weather conditions are essential for safe transit and compliance.

## Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates                  | Speed Limit |
|----|-----------|------------|------------------------------|-------------|
| 0  | U-Turn    | High       | 12.075596, 78.11078309999999 | 10 KM/Hr    |
| 1  | Turn      | High       | 11.43961, 77.87341           | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44024, 77.87528           | 15 KM/Hr    |
| 3  | Turn      | High       | 11.44898, 77.87410           | 15 KM/Hr    |
| 4  | Turn      | Medium     | 11.45357, 77.85789           | 30 KM/Hr    |
| 5  | Turn      | High       | 11.45352, 77.85698           | 15 KM/Hr    |
| 6  | Turn      | Medium     | 11.45847, 77.85140           | 30 KM/Hr    |
| 7  | Turn      | Medium     | 11.45898, 77.85135           | 30 KM/Hr    |
| 8  | Turn      | High       | 11.46303, 77.84945           | 15 KM/Hr    |
| 9  | Turn      | Medium     | 11.45543, 77.81740           | 30 KM/Hr    |
| 10 | Turn      | High       | 11.45631, 77.81668           | 15 KM/Hr    |
| 11 | Turn      | High       | 11.49383, 77.88442           | 15 KM/Hr    |
| 12 | Turn      | Medium     | 11.56083, 77.89921           | 30 KM/Hr    |
| 13 | Turn      | Medium     | 11.56180, 77.90019           | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 14 | Turn       | Medium     | 11.57048, 77.90224 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64217, 77.95200 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.64712, 77.95118 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.67858, 77.96834 | 15 KM/Hr    |
| 18 | Turn       | High       | 11.70233, 77.97869 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.71394, 77.99572 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.71916, 78.01117 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.72559, 78.01758 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.73787, 78.03528 | 30 KM/Hr    |
| 23 | Turn       | High       | 11.73651, 78.03868 | 15 KM/Hr    |
| 24 | Turn       | Medium     | 11.74120, 78.04588 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.74257, 78.04708 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.87998, 78.08170 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.91266, 78.05852 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.93746, 78.05331 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.95803, 78.04964 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.95854, 78.04990 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.96015, 78.04945 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.96370, 78.05142 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.96433, 78.05113 | 30 KM/Hr    |
| 34 | Turn       | High       | 11.96542, 78.04938 | 15 KM/Hr    |
| 35 | Turn       | High       | 11.96650, 78.04927 | 15 KM/Hr    |
| 36 | Turn       | Medium     | 11.96804, 78.05080 | 30 KM/Hr    |
| 37 | Blind Spot | Blind Spot | 12.07560, 78.11078 | 10 KM/Hr    |
| 38 | Turn       | Medium     | 12.06647, 78.10966 | 30 KM/Hr    |
| 39 | Turn       | High       | 12.06627, 78.10987 | 15 KM/Hr    |
| 40 | Turn       | High       | 12.06612, 78.10980 | 15 KM/Hr    |

### Emergency Locations

|   | type     | name                 | coordinates          | speed_limit | risk_level |
|---|----------|----------------------|----------------------|-------------|------------|
| 1 | hospital | Sri Shanthi Hospital | 11.569272, 77.901937 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 2  | hospital | Sugumar Hospital                   | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3  | hospital | Omalur Government Hospital         | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 4  | clinic   | R.R. Nursing Home                  | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 5  | hospital | Sri Hari Hospital                  | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 6  | hospital | Saravana Hospital                  | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 7  | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 8  | hospital | Sri Sugam Hospital                 | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 9  | hospital | Arun Hospital                      | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 11 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 12 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 13 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 14 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 16 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 17 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 18 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 19 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 20 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 21 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 23 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 25 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 27 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 28 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 29 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |

### Crowded Spots

|   | type   | name                       | coordinates               | speed_limit | risk_level |
|---|--------|----------------------------|---------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |



|    | type    | name                                                 | coordinates               | speed_limit | risk_level |
|----|---------|------------------------------------------------------|---------------------------|-------------|------------|
| 10 | school  | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 24 | school  | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 26 | college | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** U-Turn

**Risk Level:** High

**Speed Limit:** 10 KM/Hr

**Coordinates:** 12.075596, 78.11078309999999



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44024, 77.87528





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44898, 77.87410**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45357, 77.85789**





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45543, 77.81740





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49383, 77.88442



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56083, 77.89921





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56180, 77.90019



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57048, 77.90224





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64217, 77.95200



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64712, 77.95118



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67858, 77.96834





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70233, 77.97869



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71394, 77.99572





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71916, 78.01117



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72559, 78.01758



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73787, 78.03528





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73651, 78.03868



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.74120, 78.04588





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.74257, 78.04708**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.87998, 78.08170**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.91266, 78.05852**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.93746, 78.05331



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95803, 78.04964





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95854, 78.04990



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96015, 78.04945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96370, 78.05142





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96433, 78.05113



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.96542, 78.04938





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.96650, 78.04927



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96804, 78.05080



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 12.07560, 78.11078



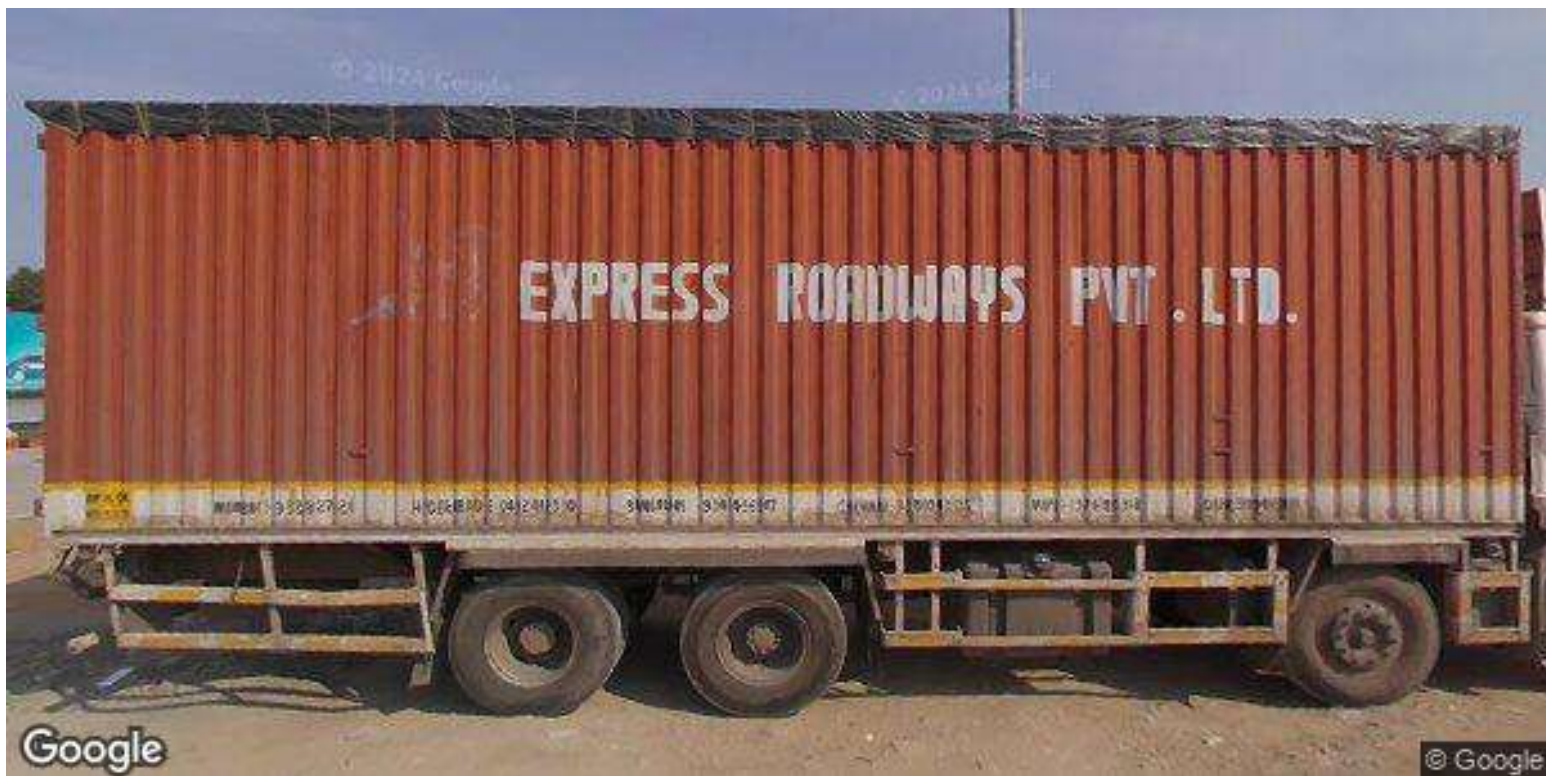


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.06647, 78.10966



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.06627, 78.10987





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.06612, 78.10980

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